

BS – 101 Engineering Physics

Lecture – 5

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Electric Potential

- When a charged particle moves in an electric field, the field exerts a force that can do *work* on the particle. This work can be expressed in terms of electric potential energy. Just as gravitational potential energy depends on the height of a mass above the earth's surface, electric potential energy depends on the position of the charged particle in the electric field.



Electric Potential

- We'll use a new concept called *electric potential*, or simply *potential*, to describe electric potential energy. In circuits, a difference in potential from one point to another is often called *voltage*. The concepts of potential and voltage are crucial to understanding how electric circuits work and have equally important applications to electron beams used in cancer radiotherapy, high-energy particle accelerators, and many other devices.

Work

- when a force $\mathbf{F}$ acts on a particle that moves from point a to point b , the work $W_{a \rightarrow b}$ done by the force is given by a *line integral*:

$$W_{a \rightarrow b} = \int_a^b \vec{F} \cdot d\vec{l} = \int_a^b F \cos \phi \, dl \quad (\text{work done by a force})$$

- where $d\mathbf{l}$ is an infinitesimal displacement along the particle's path and ϕ is the angle between $\mathbf{F}$ and $d\mathbf{l}$ at each point along the path.

Work

- Force F is *conservative*, the work done by F can always be expressed in terms of a **potential energy** U . When the particle moves from a point where the potential energy is U_a to a point where it is U_b , the change in potential energy is $\Delta U = U_b - U_a$

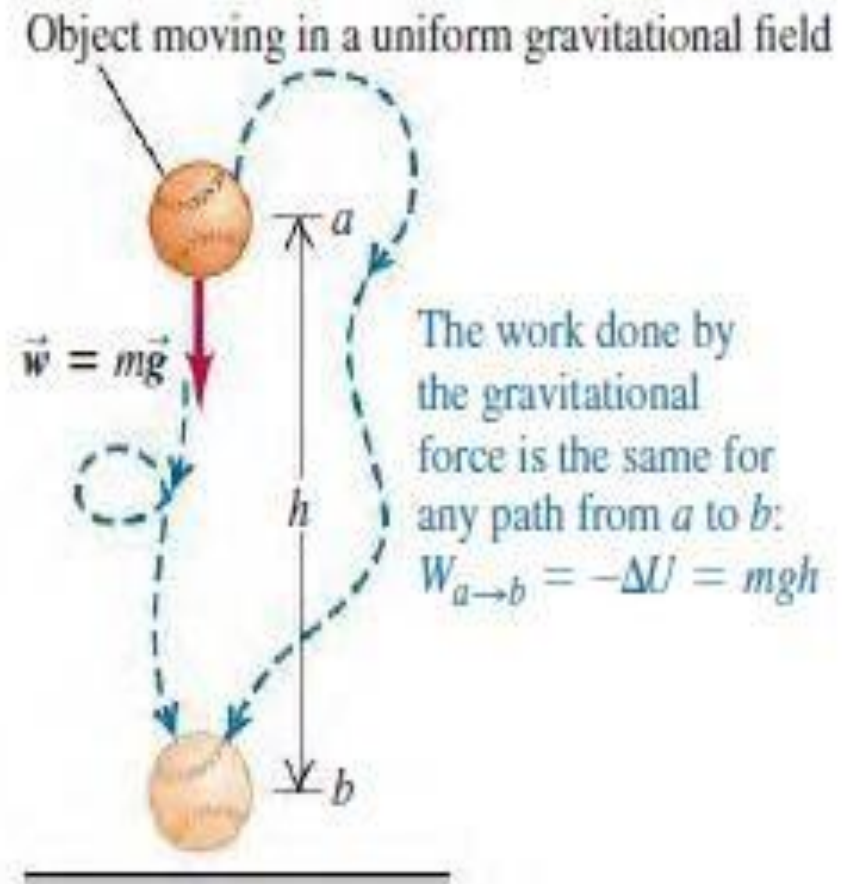
The diagram illustrates the relationship between work done by a conservative force and the change in potential energy. It features the equation $W_{a \rightarrow b} = U_a - U_b = -(U_b - U_a) = -\Delta U$. Annotations include: 'Work done by a conservative force' with an arrow pointing to $W_{a \rightarrow b}$; 'Potential energy at initial position' with an arrow pointing to U_a ; 'Potential energy at final position' with an arrow pointing to U_b ; and 'Negative of change in potential energy' with an arrow pointing to $-\Delta U$.

$$W_{a \rightarrow b} = U_a - U_b = -(U_b - U_a) = -\Delta U$$

- When $W_{a \rightarrow b}$ is positive, U_a is greater than U_b , ΔU is negative, and the potential energy *decreases*.

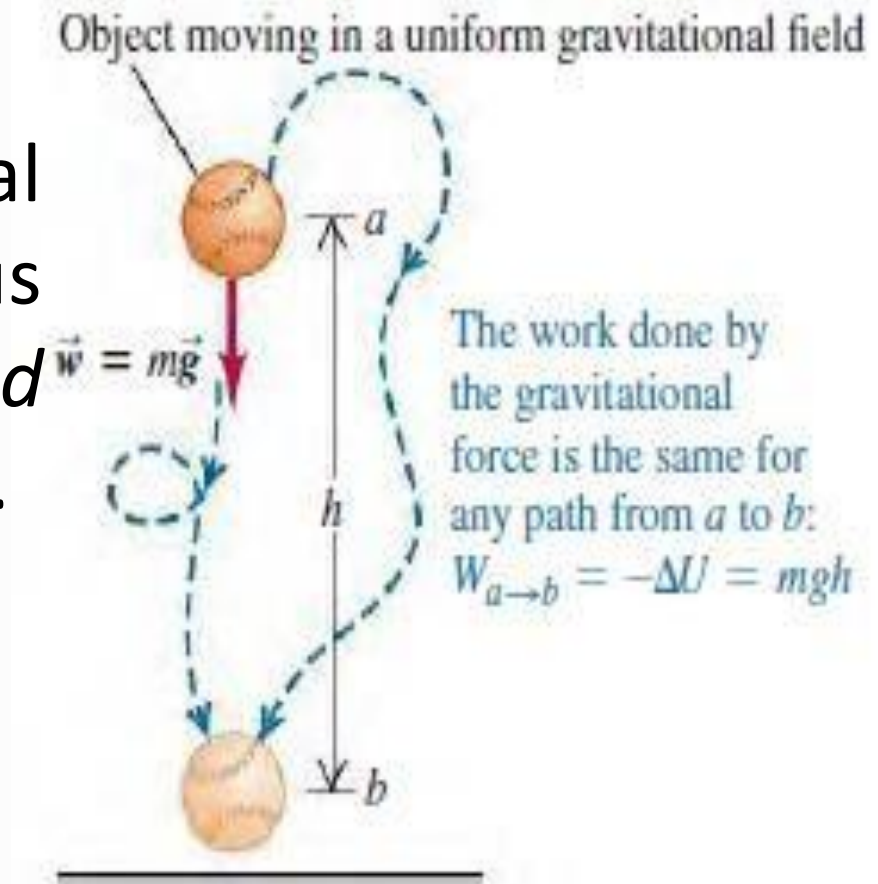
Work

- When a baseball falls from a high point (a) to a lower point (b) under the influence of the earth's gravity; the force of gravity does positive work, and the gravitational potential energy decreases. When a tossed ball is moving upward, the gravitational force does negative work during the ascent, and the potential energy increases.



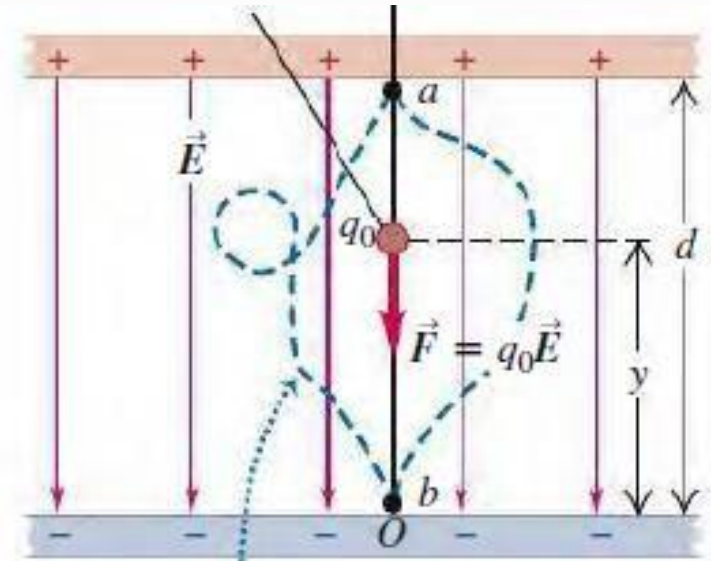
Work

- $K_a + U_a = K_b + U_b$
- The total mechanical energy (kinetic plus potential) is *conserved* under these circumstances.



Electric Potential Energy in a Uniform Field

- $W_{a \rightarrow b} = Fd$
- $E = \frac{F}{q_0}$
- $W_{a \rightarrow b} = q_0 E d$



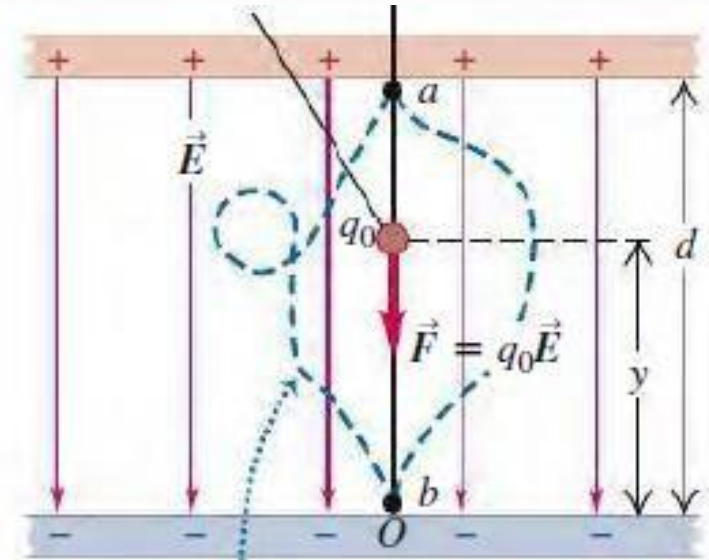
The work done by the electric force is the same for any path from a to b :

$$W_{a \rightarrow b} = -\Delta U = q_0 E d$$

- Electric Potential Energy $\rightarrow U$
- $U = q_0 E y$

Electric Potential Energy in a Uniform Field

- When the test charge moves from height y_a to height y_b , the work done on the charge by the field is given by



The work done by the electric force is the same for any path from a to b :

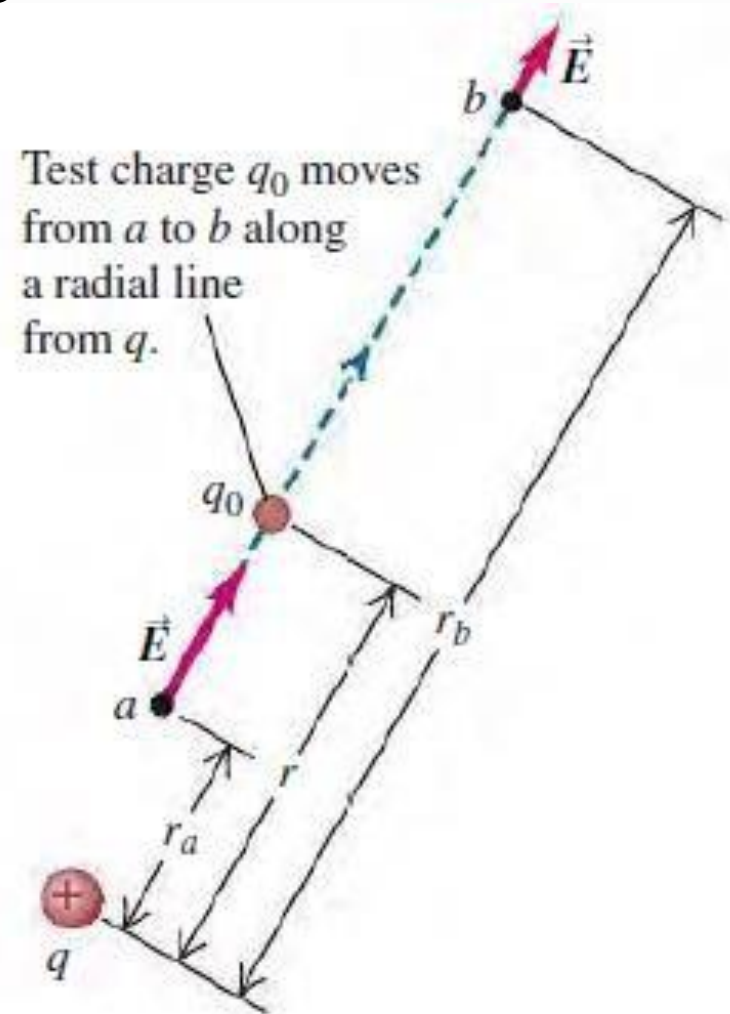
$$W_{a \rightarrow b} = -\Delta U = q_0 E d$$

$$W_{a \rightarrow b} = -\Delta U = -(U_b - U_a) = -(q_0 E y_b - q_0 E y_a) = q_0 E (y_a - y_b)$$

Electric Potential Energy of Two Point Charges

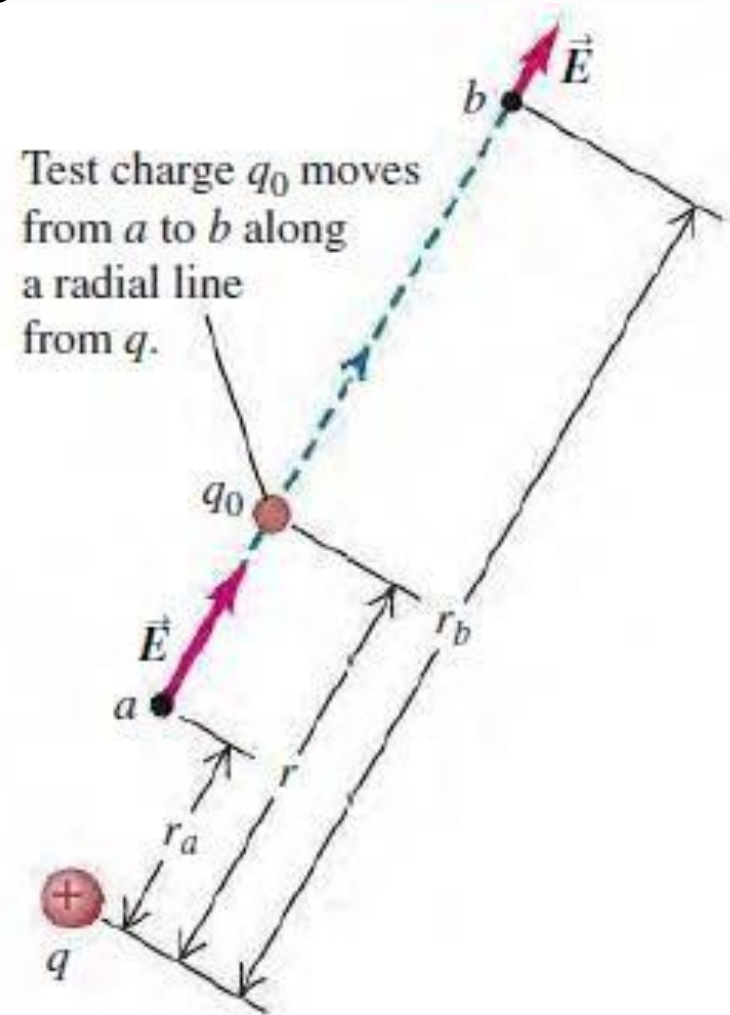
- We will calculate the work done on a test charge q_0 moving in the electric field caused by a single, stationary point charge q .
- We'll consider first a displacement along the *radial* line. The force on q_0 is given by Coulomb's law, and its radial component is

$$F_r = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2}$$



Electric Potential Energy of Two Point Charges

- If q and q_0 have the same sign (+ or -), the force is repulsive and F_r is positive; if the two charges have opposite signs, the force is attractive and F_r is negative. The force is not constant during the displacement, and we must integrate to calculate the work $W_{a \rightarrow b}$ done on q_0 by this force as q_0 moves from a to b :





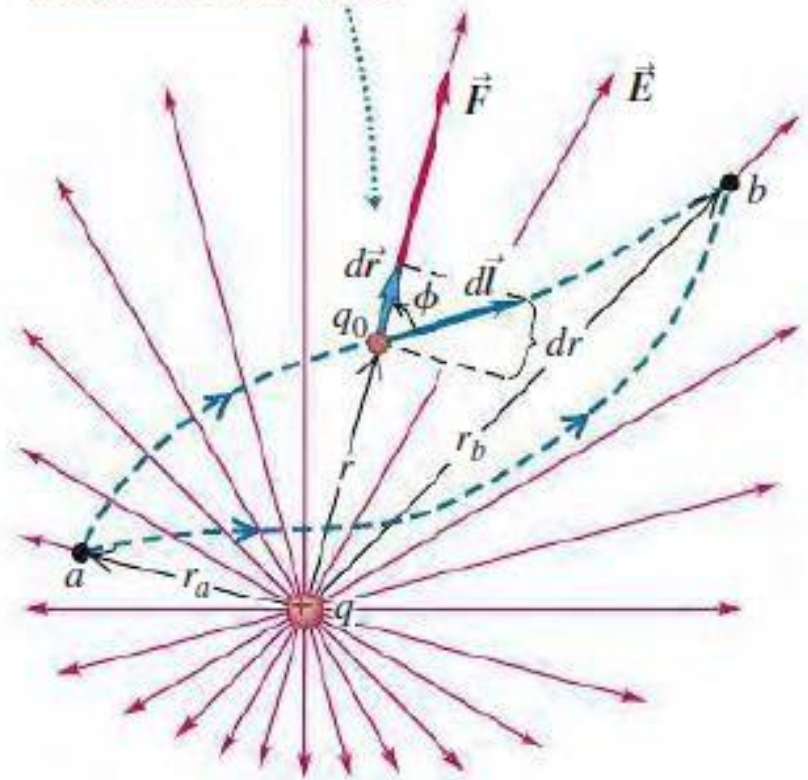
Electric Potential Energy of Two Point Charges

$$W_{a \rightarrow b} = \int_{r_a}^{r_b} F_r dr = \int_{r_a}^{r_b} \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} dr = \frac{qq_0}{4\pi\epsilon_0} \left(\frac{1}{r_a} - \frac{1}{r_b} \right)$$

Electric Potential Energy of Two Point Charges

- Now let's consider a more general displacement in which a and b do *not* lie on the same radial line. The work done on q_0 during this displacement is given by

Test charge q_0 moves from a to b along an arbitrary path.



Electric Potential Energy of Two Point Charges

$$W_{a \rightarrow b} = \int_{r_a}^{r_b} F \cos \phi \, dl = \int_{r_a}^{r_b} \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} \cos \phi \, dl$$

Electric potential energy of two point charges

$$U = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r}$$

Electric constant

Values of two charges

Distance between two charges